

Solving Quadratic Equations with the Quadratic Formula

Quadratic Equations with Complex Solutions

Solve by completing the square.

$$\textcircled{1} -3x^2 + 7x - 8 = 0$$

$$\frac{-3x^2 + 7x - 8 = 0}{-3 \quad -3}$$

$$x^2 - \frac{7}{3}x + \frac{8}{3} = 0$$

$$x^2 - \frac{7}{3}x + \frac{49}{36} = -\frac{8}{3} + \frac{49}{36}$$

$$(x - \frac{7}{6})(x - \frac{7}{6}) = -\frac{96}{36} + \frac{49}{36}$$

$$(x - \frac{7}{6})^2 = -\frac{47}{36}$$

$$\sqrt{(x - \frac{7}{6})^2} = \pm \sqrt{-\frac{47}{36}}$$

$$x - \frac{7}{6} = \pm \frac{\sqrt{-47}}{\sqrt{36}}$$

$$x - \frac{7}{6} = \pm \frac{i\sqrt{47}}{6}$$

$$x = \frac{7}{6} \pm \frac{i\sqrt{47}}{6}$$

$$\boxed{x = \frac{7 \pm i\sqrt{47}}{6}}$$

$$\textcircled{2} 0 = 5x^2 - 4x + 2$$

$$\frac{0 = 5x^2 - 4x + 2}{5 \quad 5}$$

$$0 = x^2 - \frac{4}{5}x + \frac{2}{5}$$

$$-\frac{2}{5} = x^2 - \frac{4}{5}x$$

$$\frac{4}{25} - \frac{2}{5} = x^2 - \frac{4}{5}x + \frac{4}{25}$$

$$\frac{4}{25} - \frac{10}{25} = (x - \frac{2}{5})(x - \frac{2}{5})$$

$$-\frac{6}{25} = (x - \frac{2}{5})^2$$

$$\pm \sqrt{-\frac{6}{25}} = \sqrt{(x - \frac{2}{5})^2}$$

$$\pm \frac{\sqrt{-6}}{\sqrt{25}} = x - \frac{2}{5}$$

$$\pm \frac{i\sqrt{6}}{5} = x - \frac{2}{5}$$

$$\frac{2}{5} \pm \frac{i\sqrt{6}}{5} = x \quad \boxed{x = \frac{2 \pm i\sqrt{6}}{5}}$$

$$\textcircled{3} -2x^2 + 5x - 5 = 0$$

$$\frac{-2x^2 + 5x - 5 = 0}{-2 \quad -2}$$

$$x^2 - \frac{5}{2}x + \frac{5}{2} = 0$$

$$x^2 - \frac{5}{2}x + \frac{25}{16} = -\frac{5}{2} + \frac{25}{16}$$

$$(x - \frac{5}{4})(x - \frac{5}{4}) = -\frac{40}{16} + \frac{25}{16}$$

$$(x - \frac{5}{4})^2 = -\frac{15}{16}$$

$$\sqrt{(x - \frac{5}{4})^2} = \pm \sqrt{-\frac{15}{16}}$$

$$x - \frac{5}{4} = \pm \frac{\sqrt{-15}}{\sqrt{16}}$$

$$x - \frac{5}{4} = \pm \frac{i\sqrt{15}}{4}$$

$$x = \frac{5}{4} \pm \frac{i\sqrt{15}}{4}$$

$$\boxed{x = \frac{5 \pm i\sqrt{15}}{4}}$$

$$\textcircled{4} 0 = 4x^2 - 3x + 3$$

$$\frac{0 = 4x^2 - 3x + 3}{4 \quad 4}$$

$$0 = x^2 - \frac{3}{4}x + \frac{3}{4}$$

$$-\frac{3}{4} = x^2 - \frac{3}{4}x$$

$$\frac{9}{64} - \frac{3}{4} = x^2 - \frac{3}{4}x + \frac{9}{64}$$

$$\frac{9}{64} - \frac{48}{64} = (x - \frac{3}{8})(x - \frac{3}{8})$$

$$-\frac{39}{64} = (x - \frac{3}{8})^2$$

$$\pm \sqrt{-\frac{39}{64}} = \sqrt{(x - \frac{3}{8})^2}$$

$$\pm \frac{\sqrt{-39}}{\sqrt{64}} = x - \frac{3}{8}$$

$$\pm \frac{i\sqrt{39}}{8} = x - \frac{3}{8}$$

$$\frac{3}{8} \pm \frac{i\sqrt{39}}{8} = x$$

$$\boxed{\frac{3 \pm i\sqrt{39}}{8} = x}$$

$$⑤ \quad 6x^2 - x + 2 = 0$$

$$\frac{6x^2 - x + 2}{6} = \frac{0}{6}$$

$$x^2 - \frac{1}{6}x + \frac{2}{6} = 0$$

$$x^2 - \frac{1}{6}x + \frac{1}{3} = 0$$

$$x^2 - \frac{1}{6}x = -\frac{1}{3}$$

$$x^2 - \frac{1}{6}x + \frac{1}{144} = -\frac{1}{3} + \frac{1}{144}$$

$$(x - \frac{1}{12})(x - \frac{1}{12}) = -\frac{48}{144} + \frac{1}{144}$$

$$(x - \frac{1}{12})^2 = \frac{-47}{144}$$

$$\sqrt{(x - \frac{1}{12})^2} = \pm \sqrt{\frac{-47}{144}}$$

$$x - \frac{1}{12} = \pm \frac{\sqrt{-47}}{\sqrt{144}}$$

$$x - \frac{1}{12} = \pm \frac{i\sqrt{47}}{12}$$

$$x = \frac{1}{12} \pm \frac{i\sqrt{47}}{12}$$

$$x = \frac{1 \pm i\sqrt{47}}{12}$$

$$⑥ \quad 0 = 8x^2 + 4x + 3$$

$$\frac{0}{8} = \frac{8x^2 + 4x + 3}{8}$$

$$0 = x^2 + \frac{4}{8}x + \frac{3}{8}$$

$$0 = x^2 + \frac{1}{2}x + \frac{3}{8}$$

$$-\frac{3}{8} = x^2 + \frac{1}{2}x$$

$$\frac{1}{16} - \frac{3}{8} = x^2 + \frac{1}{2}x + \frac{1}{16}$$

$$\frac{1}{16} - \frac{6}{16} = (x + \frac{1}{4})(x + \frac{1}{4})$$

$$-\frac{5}{16} = (x + \frac{1}{4})^2$$

$$\pm \sqrt{\frac{-5}{16}} = \sqrt{(x + \frac{1}{4})^2}$$

$$\pm \frac{\sqrt{-5}}{\sqrt{16}} = x + \frac{1}{4}$$

$$\pm \frac{i\sqrt{5}}{4} = x + \frac{1}{4}$$

$$-\frac{1}{4} \pm \frac{i\sqrt{5}}{4} = x \quad \rightarrow \quad x = \frac{-1 \pm i\sqrt{5}}{4}$$

The Quadratic Formula

Let's find a formula that we can use to solve any quadratic equation. We will start with a generic quadratic equation and then solve by completing the square.

$$ax^2 + bx + c = 0$$

$$\frac{ax^2 + bx + c}{a} = \frac{0}{a}$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

$$x^2 + \frac{b}{a}x = -\frac{c}{a}$$

$$x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = -\frac{c}{a} + \frac{b^2}{4a^2}$$

$$(x + \frac{b}{2a})(x + \frac{b}{2a}) = \frac{-4ac}{4a^2} + \frac{b^2}{4a^2}$$

$$(x + \frac{b}{2a})^2 = \frac{-4ac + b^2}{4a^2}$$

$$\sqrt{(x + \frac{b}{2a})^2} = \pm \sqrt{\frac{-4ac + b^2}{4a^2}}$$

$$x + \frac{b}{2a} = \pm \frac{\sqrt{-4ac + b^2}}{\sqrt{4a^2}}$$

$$x + \frac{b}{2a} = \pm \frac{\sqrt{-4ac + b^2}}{2a}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{I) } \frac{b}{a} \div \frac{2}{1} = \frac{b}{a} \cdot \frac{1}{2} = \frac{b}{2a}$$

$$\text{II) } \left(\frac{b}{2a}\right)^2 = \frac{b}{2a} \cdot \frac{b}{2a} = \frac{b^2}{4a^2}$$

Solve using the quadratic formula.

$$\textcircled{7} 3x^2 + 5x + 1 = 0$$

$$ax^2 + bx + c = 0$$

$$a=3, b=5, c=1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-5 \pm \sqrt{5^2 - 4(3)(1)}}{2(3)}$$

$$x = \frac{-5 \pm \sqrt{25 - 12}}{6}$$

$$x = \frac{-5 \pm \sqrt{13}}{6}$$

$$\textcircled{9} x^2 - 9x + 9 = 0$$

$$a=1, b=-9, c=9$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-9) \pm \sqrt{(-9)^2 - 4(1)(9)}}{2(1)}$$

$$\textcircled{8} 0 = -7x^2 + 3x + 2$$

$$a=-7, b=3, c=2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4(-7)(2)}}{2(-7)}$$

$$x = \frac{-3 \pm \sqrt{9 + 56}}{-14}$$

$$x = \frac{-3 \pm \sqrt{65}}{-14}$$

$$x = \frac{3 \pm \sqrt{65}}{14}$$

$$x = \frac{9 \pm \sqrt{81 - 36}}{2}$$

$$x = \frac{9 \pm \sqrt{45}}{2}$$

$$x = \frac{9 \pm \sqrt{9 \cdot 5}}{2}$$

$$x = \frac{9 \pm 3\sqrt{5}}{2}$$

$$\textcircled{10} 6x^2 - 2x - 3 = 0$$

$$a=6, b=-2, c=-3$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(6)(-3)}}{2(6)}$$

$$x = \frac{2 \pm \sqrt{4 + 72}}{12}$$

$$x = \frac{2 \pm \sqrt{76}}{12}$$

$$x = \frac{2 \pm 2\sqrt{19}}{12}$$

$$x = \frac{1 \pm \sqrt{19}}{6}$$

$$\textcircled{12} 0 = -x^2 + 3x + 7$$

$$a=-1, b=3, c=7$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4(-1)(7)}}{2(-1)}$$

$$\textcircled{11} -6x^2 + 5x + 3 = 0$$

$$a=-6, b=5, c=3$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-5 \pm \sqrt{5^2 - 4(-6)(3)}}{2(-6)}$$

$$x = \frac{-5 \pm \sqrt{25 + 72}}{-12}$$

$$x = \frac{-5 \pm \sqrt{97}}{-12}$$

$$x = \frac{5 \pm \sqrt{97}}{12}$$

$$x = \frac{-3 \pm \sqrt{9 + 28}}{-2}$$

$$x = \frac{-3 \pm \sqrt{37}}{-2}$$

$$x = \frac{3 \pm \sqrt{37}}{2}$$

$$(13) 0 = -3x^2 + 6x - 1$$

$$a = -3, b = 6, c = -1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-6 \pm \sqrt{6^2 - 4(-3)(-1)}}{2(-3)}$$

$$x = \frac{-6 \pm \sqrt{36 - 12}}{-6}$$

$$x = \frac{-6 \pm \sqrt{24}}{-6}$$

$$x = \frac{-6 \pm 2\sqrt{6}}{-6}$$

$$x = \frac{3 \pm \sqrt{6}}{3}$$

$$(15) 0 = -2x^2 - 3x + 5$$

$$a = -2, b = -3, c = 5$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(-2)(5)}}{2(-2)}$$

$$(14) 11x^2 + x - 1 = 0$$

$$a = 11, b = 1, c = -1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-1 \pm \sqrt{1^2 - 4(11)(-1)}}{2(11)}$$

$$x = \frac{-1 \pm \sqrt{1 + 44}}{22}$$

$$x = \frac{-1 \pm \sqrt{45}}{22}$$

$$x = \frac{-1 \pm 3\sqrt{5}}{22}$$

$$x = \frac{3 \pm \sqrt{9 + 40}}{-4}$$

$$x = \frac{3 \pm \sqrt{49}}{-4}$$

$$x = \frac{3 \pm 7}{-4}$$

$$x = -5/2 \text{ or } x = 1$$

$$(16) x^2 + 8x + 16 = 0$$

$$a = 1, b = 8, c = 16$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-8 \pm \sqrt{8^2 - 4(1)(16)}}{2(1)}$$

$$x = \frac{-8 \pm \sqrt{64 - 64}}{2}$$

$$x = \frac{-8 \pm \sqrt{0}}{2}$$

$$x = \frac{-8 \pm 0}{2}$$

$$x = \frac{-8}{2} = -4$$

$$(18) 9x^2 + 30x + 25 = 0$$

$$a = 9, b = 30, c = 25$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-30 \pm \sqrt{30^2 - 4(9)(25)}}{2(9)}$$

$$x = \frac{-30 \pm \sqrt{900 - 900}}{18}$$

$$(17) 0 = 3x^2 - 8x - 3$$

$$a = 3, b = -8, c = -3$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4(3)(-3)}}{2(3)}$$

$$x = \frac{8 \pm \sqrt{64 + 36}}{6}$$

$$x = \frac{8 \pm \sqrt{100}}{6}$$

$$x = \frac{8 \pm 10}{6}$$

$$x = 3 \text{ or } x = -1/3$$

$$x = \frac{-30 \pm \sqrt{0}}{18}$$

$$x = \frac{-30 \pm 0}{18}$$

$$x = \frac{-30}{18}$$

$$x = -5/3$$

$$\textcircled{19} 5x^2 + 7x + 3 = 0$$

$$a=5, b=7, c=3$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-7 \pm \sqrt{7^2 - 4(5)(3)}}{2(5)}$$

$$x = \frac{-7 \pm \sqrt{49 - 60}}{10}$$

$$x = \frac{-7 \pm \sqrt{-11}}{10}$$

$$x = \frac{-7 \pm i\sqrt{11}}{10}$$

$$\textcircled{21} x^2 - 6x + 18 = 0$$

$$a=1, b=-6, c=18$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(18)}}{2(1)}$$

$$\textcircled{20} 0 = -3x^2 + 2x - 2$$

$$a=3, b=2, c=-2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4(-3)(-2)}}{2(-3)}$$

$$x = \frac{-2 \pm \sqrt{4 - 24}}{-6}$$

$$x = \frac{-2 \pm \sqrt{-20}}{-6}$$

$$x = \frac{-2 \pm \sqrt{(-1)(4)(5)}}{-6}$$

$$x = \frac{-2 \pm 2i\sqrt{5}}{-6}$$

$$x = \frac{1 \pm i\sqrt{5}}{3}$$

$$x = \frac{6 \pm \sqrt{36 - 72}}{2}$$

$$x = \frac{6 \pm \sqrt{-36}}{2}$$

$$x = \frac{6 \pm 6i}{2} = 3 \pm 3i$$

$$\textcircled{22} -3x^2 + 7x - 6 = 0$$

$$a=-3, b=7, c=-6$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-7 \pm \sqrt{7^2 - 4(-3)(-6)}}{2(-3)}$$

$$x = \frac{-7 \pm \sqrt{49 - 72}}{-6}$$

$$x = \frac{-7 \pm \sqrt{-23}}{-6}$$

$$x = \frac{7 \pm i\sqrt{23}}{6}$$

$$\textcircled{23} 0 = 7x^2 - x + 1$$

$$a=7, b=-1, c=1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(7)(1)}}{2(7)}$$

$$x = \frac{1 \pm \sqrt{1 - 28}}{14}$$

$$x = \frac{1 \pm \sqrt{-27}}{14}$$

$$x = \frac{1 \pm 3i\sqrt{3}}{14}$$

$$\textcircled{24} x^2 - 2x + 2 = 0$$

$$a=1, b=-2, c=2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(2)}}{2(1)}$$

$$x = \frac{2 \pm \sqrt{4 - 8}}{2}$$

$$x = \frac{2 \pm \sqrt{-4}}{2}$$

$$x = \frac{2 \pm 2i}{2}$$

$$x = 1 \pm i$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The Discriminant

$$D = b^2 - 4ac$$

The discriminant gives us information about the number of solutions of a quadratic equation and the type of solutions because it lies under the root symbol in the quadratic formula.

I) If the discriminant of a quadratic equation is **positive**, then the equation has **two real solutions**.

II) If the discriminant of a quadratic equation is **zero**, then the equation has **one real solution**.

III) If the discriminant of a quadratic equation is **negative**, then the equation has **two complex solutions**.

Find the discriminant of each of the following quadratic equations and use it to determine the number and type of solutions. Write "one real solution", "two real solutions", or "two complex solutions".

$$(25) \quad 3x^2 + 5x - 2 = 0 \quad D = 5^2 - 4(3)(-2)$$

$$a = 3, b = 5, c = -2 \quad D = 25 + 24$$

$$D = b^2 - 4ac$$

$$D = 49$$

Two real solutions

$$(26) \quad 2x^2 - x + 1 = 0$$

$$a = 2, b = -1, c = 1$$

$$D = b^2 - 4ac$$

$$D = (-1)^2 - 4(2)(1)$$

$$D = 1 - 8$$

$$D = -7$$

Two complex solutions

$$(28) \quad -5x^2 + 2x - 3 = 0$$

$$a = -5, b = 2, c = -3$$

$$D = b^2 - 4ac$$

$$D = 2^2 - 4(-5)(-3)$$

$$D = 4 - 60$$

$$D = -56$$

Two complex solutions

$$(30) \quad 9x^2 + 12x + 4 = 0$$

$$a = 9, b = 12, c = 4$$

$$D = b^2 - 4ac$$

$$(27) \quad 0 = -x^2 + 4x - 4$$

$$a = -1, b = 4, c = -4$$

$$D = b^2 - 4ac$$

$$D = 4^2 - 4(-1)(-4)$$

$$D = 16 - 16$$

$$D = 0$$

One real solution

$$(29) \quad 0 = x^2 + 7x + 10$$

$$a = 1, b = 7, c = 10$$

$$D = b^2 - 4ac$$

$$D = 7^2 - 4(1)(10)$$

$$D = 49 - 40$$

$$D = 9$$

Two real solutions

$$D = 12^2 - 4(9)(4)$$

$$D = 144 - 144$$

$$D = 0$$

One real solution